

biochemistry paper 2 notes - enzyme catalysis

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August 2026

1 Transition State Theory

Transition state happens at the highest energy point of a reaction.

Derivation of reaction rate: transition state and ground state are in thermodynamic equilibrium. Thus, calculable via energy difference. For a unimolecular reaction: Consider: $X^\ddagger$, the transition state. X , the ground state and $\Delta G^\ddagger$, the energy difference between them. Then:

$$[X^\ddagger] = [X] \exp\left(-\frac{\Delta G^\ddagger}{RT}\right)$$

Transition state decomposition rate = breaking bond vibration frequency ν . From the energy of an excited oscillator $E = h\nu$ and $E = kT$:

$$\nu = \frac{kT}{h}$$

rate of decomposition:

$$-\frac{d[X]}{dt} = \nu[X^\ddagger] = [X]\left(\frac{kT}{h}\right) \exp\left(-\frac{\Delta G}{RT}\right)$$

used in comparing the relative reactivities of pairs of substrates.

1.1 Principles of Catalysis

The uncatalyzed attack of water on an ester leads to a transition state leading to the formation of a positive charge on the water's oxygen.

The hydrolysis of an acetal leads to a transition state with structure similar to a carbonium ion and an alkoxide ion. Develops a negative charge $\rightarrow$ transition state is unstable.

General base catalysis (first reaction): reaction is catalyzed via transition state energy lowering: positive charge on the attacking water molecule by transferring one of the protons to a base

General acid catalysis (second reaction): protonation of an expelled alcohol from an acid $\rightarrow$ stabilization.

Electrophilic catalysis: stabilization of a negative charge via a metal ion (Zn^{2+} or Mg^{2+}).

Nucleophilic catalysis: catalysis where the nucleophile is more nucleophilic than the one it replaces, intermediate must be more reactive than the parent compound.

2 glycosyl transfer catalyzed by enzymes

2.1 Types of glycosyl transfer

Types of glycosyl transfer:

- first mechanism (ping-pong): one substrate is bound to transformation (often with loss of molecular fragment) while the second substrate binds in, releasing the product.
- ternary complex mechanism: both substrates bind at the active site before any catalysis. Called sequential due to sequential substrate binding.

Glycosidase: enzyme responsible to transferring glycosyl to water. Happens with either inversion or retention of configuration. In the majority of the **retaining glycoside hydrolases**, nucleophile X is a carboxylate group from an **Asp or Glu** residue of the protein (high pH value, proton donor).

Glycosyl-enzyme intermediates in double displacement glycosidases = transglycosylases and act by pingpong mechanism.

Glycosyl transferases: use activated glycosyl donors and the nucleophilic displacement involves departure of a **phosphate** or **pyrophosphate ester** - Leloir pathway.

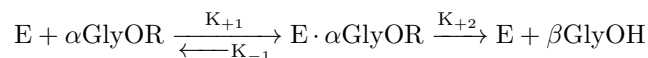
2.2 Stereochemistry and Steady-State Kinetics of Enzymatic Glycosyl Transfer

If one substrate is involved and the active sites are independent, the Michaelis-Menten equation holds:

$$\nu = \frac{k_{cat}[E]_0[S]}{K_m + [S]} = \frac{V_{max}[S]}{K_m + [S]}$$

where E_0 is the **total concentration of enzyme** and ν the **rate of reaction** at a substrate concentration $[S]$.

Minimal kinetic scheme for an inverting glycosidase:



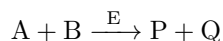
where K_{+1} is the **bimolecular rate constant** for enzyme-substrate combination, K_{-1} is the **unimolecular rate constant** for loss of the substrate from

the enzyme and K_{+2} is the **first order rate constant** for the chemical step in the ES complex.

Glycosidase inhibition failing points: inhibitor impurity, enzyme impurity, wrong measurement Competitive inhibition: competition of active site between the inhibitor and the substrate. since V_{max} is constant, all plots must intersect at that point. Anti-competitive: lines are parallel inhibitor binds only to ES complex, V_{max} decreases (ES trapped by inhibitor).

Derivation of Bloomfield-Alberty equation:

Consider a general bi-bi reaction:



with enzyme complexes: $E, EA, EB, EAB, EP, EQ, EPQ$ written in the form valid for any number of enzyme complexes under steady-state conditions.

Assume for every intermediate X :

$$\frac{d[X]}{dt} \approx 0$$

yielding a system of linear equations relating the concentrations of all enzyme species that is collapsible into single expression:

$$v = \frac{V_{max}[A][B]}{D}$$

where D is a denominator containing all binding terms.

The denominator must have the form:

$$D = \sum_i c_i \Pi_j [X_j]$$

where:

- each c_i is an observable kinetic parameter;
- each term is a product of reactant / product concentrations;
- the structure is independent of the number of intermediates

For a simple ordered bi-bi mechanism, the denominator reduces to:

$$D = K_A K_B + K_B[A] + K_A[B] + [A][B]$$

Thus,

$$v = \frac{V_{max}[A][B]}{K_A K_B + K_B[A] + K_A[B] + [A][B]}$$

yielding the Bloomfield-Alberty bi-bi rate equation.

To obtain a Lineweaver-Burk type linear form, fix one substrate and vary the other. Define

$$\alpha = 1 + \frac{K_B}{[B]}$$

with [B] fixed. Then, the law collapses to:

$$v = \frac{V_{max}[A]}{\alpha K_A + [A]}$$

Inverting yields:

$$\frac{1}{v} = \frac{\alpha K_A}{V_{max}} \frac{1}{[S]} + \frac{\alpha}{V_{max}}$$

where:

- α encodes the presence of the second substrate or inhibitor;
- slope is:

$$\frac{\alpha K_m}{V_{max}}$$

;

- intercept is:

$$\frac{\alpha}{V_{max}}$$

Stereochemistry of glycosidases and transglycosidases is determined by reactant / product structure. If no transfer products, chemical flux must be different from the mutarotation equilibrium and analytical method faster than mutarotation. Polarimetry = classical determination method: adding sodium to see reaction.

isotopic substitution = no perturbation on energy functions, good for enzyme mechanism determination. Always done in competitive inhibition, useful method of determining isotope effects = equilibrium perturbation. If presence of effect, reverse reaction will be slower.

Intrinsic binding energy: Maximum free energy of binding of a small molecule to a protein. Removing a remote transition-state interaction = increase in activation free energy $\Delta G^\ddagger \rightarrow$ drop in $k \rightarrow$ change in catalytic strategy, where k is the rate of mutation.

Exo vs Endo: Exo = non-reducing action on single sugar residue, endo = reaction in the middle of polysaccharide chain with any number of positive / negative sites.

Three types of active sites on polysaccharides: ends of chain = pits, middle of chain = clefts, crystalline / insoluble polysaccharides = tunnels removing sugar chains. multiple active sites, one saccharide residue per site. Substitution = subsite -1, leaving group = subsite +1.

Reaction of retaining glycosidases with glycals: intermediate 2-deoxyglycosyl enzyme is formed though an E2 reaction. If retaining = α -glycosidase, proton added to β -face of C2. Otherwise, to the α -phase. Reaction requires **oxocarbenium alignment**: enzyme must align the



ring oxygen with the empty p-orbital forming at C1. Protonation at C2 must happen in way that pushes the electron density towards C1 in the correct direction. β -face protonation pushes electron density downwards, aligning developing cation with the ring oxygen. α -protonation pushes upward, yielding correct oxocarbenium geometry.

Reaction of glycosidases with exocyclic enol ethers: double bond protonated by acid-catalyst; resulting tertiary cation is captured by the nucleophile of retaining glycosidases or by the water molecule of inverting ones. If inverting enzyme has subsite +1, then transfer product has opposite stereochemistry to the original substrate.

Circe effect: process of an enzyme using strong electrostatic forces and binding energy to physically pull a substrate out of the solution and trap it in active site. Enzyme utilizing Circe effect have active sites are engineered with highly specialized, charged amino-acid residues creating a ground-state vector field. Vector field exerts an attractive, pulling substrate toward the binding pocket. Enzyme uses binding energy to freeze substrate and destabilize its bonds for it to look like the high-energy transition state.

Inversion mechanisms: Inversion N-glycosidases $\rightarrow$ single-step SN2-like displacement at anomeric carbon: protonating leaving group + nucleophile activation $\rightarrow$ configuration inversion of sugar.

3 Core Geometry

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3.1 SN2-Geometry

- **Backside attack** - nucleophile approaches C1 from opposite side of leaving group;
- **Pentavalent TS** - anomeric carbon through trigonal-bipyramidal transition state;
- **Direct displacement**
- **Configuration Inversion** - $\alpha \rightarrow \beta$ or $\beta \rightarrow \alpha$ depending on substrate

3.2 Catalytic Residues

Inversion glycosidases use **two residues**, without covalent intermediate (unlike retaining enzymes)

- **General acid** - protonates leaving group (glycosidic oxygen);
- **General base** - deprotonates incoming nucleophile (water);
- **Synchronous action** - both residues act during the same step;

- **No intermediate** - unlike 2-deoxyglycosyl enzyme in retaining glycosidases

3.3 Transition State Shape

- **Associative TS** - nucleophile, leaving group partially bonded;
- **Pentavalent carbon** - C1 expands coordination temporarily;
- **Orbital alignment** - nucleophile must align with the σ orbital of C—O bond

Uracil DNA glycosidase: DNA repair enzyme, removes non-DNA base uracil from double-strand DNA. Initial product of DNA-glycosylase = β -phosphaldehyde (subjected to loss of phosphate) - accelerated by lysine.

Mechanism = Schiff base intermediate (reduced with tritiated sodium borohydride) + Amadori rearrangement to enamine, expulsion of phosphate.

- **Lysine attack** - Active-site Lys attacks C1 of desoxyribose;
- **Imine formation** - water leaves, forming a Schiff base (C=N)
- **Base release** - damaged base is expelled

Amadori rearrangement:

- 1,2-shift - imine $\rightarrow$ enamine by tautomerization;
- **Stabilization** - enamine = more stable;
- Sugar reorganized - ribose ring opens and rearranges
- **β -elimination: enamine collapses, electron towards phosphate**
- **Phosphate leaving group** forms
- **Enzyme regeneration** - Schiff base breaks by hydrolysis, releasing sugar fragments

Stereoelectronic requirements for Amadori rearrangement:

- C—H bond must align with C=N π
- Hyperconjugation in π : electrons from the C—H σ bond donate into imine's π orbital.

4 ADP-Ribosylation on Arginine

- Mono-ADP ribosylation - Enzyme transfers ADP-ribose from NAD^+ to arginine residue, forming an N-linked ADP-ribosyl-arginine;
- Arginine specific mono-ADP-ribosyltransferase (ART) - catalyzes reaction (ART1 in humans or bacterial iota-toxin)
- NAD^+ is cleaved into nicotinamide and ADP-ribose

4.1 Mechanism

- **Activation of NAD^+** - enzyme cleaves the N-glycosyl bond with NAD in a specific position;
- Nicotinamide leaving group generates an ADP-ribosyl cation
- Guanidino group of arginine attacks ribose C1, forming an N-glycosidic linkage

Suicide substrates = not overtly reactive + relies on enzyme to produce electrophilic species. If species has no recognition ability, diffuse freely. Quinone methide diffuse to attack several active sites of enzyme but does not live long enough to attack other proteins.

In paracatalytic inactivation of glycosidases, destabilizing the second transition state = slow turnover of glycosyl enzyme

CBMs (Carbohydrate binding modules): modular protein domains that bind carbohydrates without catalyzing reactions. Aromatic residues (Trp, Tyr, Phe) to stack against sugar rings. Polysaccharide oriented such that CBMs hit the glycosidic bond. Flexible links = enzyme reach and mobility.

CBM structure: flat or shallow grooves with aromatic residues for sugar binding. Trp / Tyr / Phe stack against hydrophobic phase of glucose. Backbone NH/CO groups form directional H-bonds with hydroxyls.

CBMs solve insolubility problem:

- Targets insoluble substrate: binding to surface of polysaccharide;
- Increasing local concentration: holding enzyme near substrate for long time;
- orientation of chain

CBM families:

- Type A: bind crystalline surfaces (Flat hydrophobic surfaces, strong aromatic stacking, cellulose);
- Type B: bind internal regions of single polysaccharide chains (groove binding sites, used by hemicellulases)
- Type C: bind small soluble sugars (pocket binding sites, enzymes acting on soluble sugars)

Hydrophobic effect of Type A CBMs: caused by ordering of water molecules around a non-polar solute to form tighter hydrogen bonds. Function = driving force is hydrophobic $\rightarrow$ Type A binds to hydrophobic faces of polysaccharide chain.

CBM binding with three tyrosines is reversible. Changing one to tryptophan made binding irreversible. Conversion of binding face from trp to tyr produced a reversible CBD binding.

Type B CBM depend on interactions of planar aromatic residues + hydrogen bonding interactions.

example of type B bonding: chloroform-benzene interaction: mutation of hydrogen bonding = no effect on overall affinity but changed the entropic / enthalpic contribution to binding (entropic more likely)

Type C binding: more dependent on hydrogen bonding and enthalpic-driven.

5 Retaining N-Glycosylases

Mechanism:

- N-glycosidic bond cleavage: enzyme protonates nicotinamide giving a ribosyl cation;
- Ribose adopts same TS geometry as retaining glycosidases;
- Front-side attack = retaining mechanism

Mechanism: water attacks ribosyl cation, front side attack = retaining stereochemistry, ADP-ribose released, roles: NAD turnover, signaling, nicotinamide release.

NAD cyclases: same ribosyl cation as the glycosylase but folded back onto adenine ring.

Mechanism: ribosyl cation attacked by adenine nitrogen, forms a new N-glycosidic bond, retention mechanism.

6 tRNA Transglycosylases

Replaces the first synthesized guanine residues in tRNA for unusual bases, nucleophile = Asp

2-Desoxyribosyl Transferase:

- Class I: accepts only purines;
- Class II: accepts both purines and pyrimidines;
- Nucleophile is a glutamate via Withers inactivator
- Acid-base machinery is related to base specificity and presence of protein loops.

Glycosyl transferase:

- A fold = N-terminal nucleotide binding (four parallel β strands flanked on both sides by two α helices $\rightarrow$ disordered regions of β sheets and α helices, foldable over glycosyl;
- B fold = active site between two Rossmann domains, function = catalyze transfer of glycosyl from activated donors by retention or inversion

References

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